

Sample	Good Index 1								Good Index 2									
1	701	T	A	A	G	G	C	G	A	702	C	G	T	A	C	T	A	C
2	702	C	G	T	A	C	T	A	G	703	A	G	G	C	A	G	A	A
3	704	T	C	C	T	G	A	G	C	711	A	A	G	A	G	G	C	A
		✓	✓	✓	✓	✓	✓	✓	✓	X	✓	X	X	✓	X	X	✓	

✓= signal in both colors
X= signal missing in 1 color channel

Good Examples								Bad Examples																											
Index 1				Index 2				Index 1				Index 2																							
705	G	G	A	C	T	C	C	T	503	T	A	T	C	C	T	C	T	705	G	G	A	C	T	C	C	T	502	C	T	C	T	C	T	A	T
706	T	A	G	G	C	A	T	G	503	T	A	T	C	C	T	C	T	706	T	A	G	G	C	A	T	G	502	C	T	C	T	C	T	A	T
701	T	A	A	G	G	C	G	A	504	A	G	A	G	T	A	G	A	701	T	A	A	G	G	C	G	A	503	T	A	T	C	C	T	C	T
702	C	G	T	A	C	T	A	G	504	A	G	A	G	T	A	G	A	702	C	G	T	A	C	T	A	G	503	T	A	T	C	C	T	C	T
✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓				✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓				✓ ✓ ✓ ✓ ✓ ✓ ✓ ✓				✓ ✓ ✓ ✓ X X X X																							
✓ = signal in both colors																																			
X = signal missing in 1 color channel																																			

NextSeq Systems use 2-channel sequencing, which requires only 2 images to encode the data for 4 DNA bases: 1 red channel and 1 green channel. The NextSeq also uses a new implementation of Real-Time Analysis (RTA) called RTA2.0, which includes important architecture differences from RTA on other Illumina sequencing systems. For any index sequences, RTA2.0 requires that there is at least one base other than G in the first 2 cycles. This requirement for index diversity allows the use of any Illumina index selection for single-plex indexing except index 1 (i7) 705, which uses the sequence GGACTCCT. Use the combinations in Table 6 for proper color balancing on NextSeq Systems.

Plexity	Index 1 (i7) Selection	Index 2 (i5) Selection	
3 plex	N702, N704, N705	Fixed Index 2 adapter	
6 plex	N701, N702, N703, N704, N705, N707		
9 plex	N701, N702, N703, N704, N705, N707, N710, N711, N712		
12 plex	N701, N702, N703, N704, N705, N706, N707, N708, N709, N710, N711, N712		
When pooling:			
a. 3–6 plex—Use the indexes required for a 3-plex pool and select from the indexes required for a 6-plex pool.			
b. 6–9 plex—Use the indexes required for a 6-plex pool and select from the indexes required for a 9-plex pool.			
c. 9–12 plex—Use the indexes required for a 9-plex pool and select from the indexes required for a 12-plex pool.			